Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 — 2024

Data: Global Carbon Budget 2025 · NASA · Our World in Data

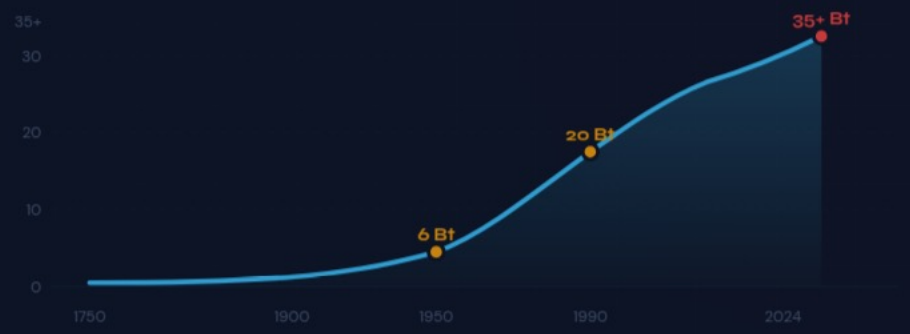
01 Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429	280	+53%
ppm (Feb 2026)	ppm pre-industrial	increase since 1750

Source: NOAA Mauna Loa Observatory / NASA

02 Global Fossil Fuel CO_2 Grew Six-Fold: 6 Bt → 35+ Bt

EMISSIONS TRAJECTORY (1750–2024)



1950
6 billion tonnes

1990
20 billion tonnes

2024
35+ billion tonnes

03 China Now Emits 12.5 Bt — More Than Double the United States

COUNTRY / REGION	ANNUAL CO_2 (BT)	SHARE	TREND
China	~12.5	~27%	▲ Slight increase
United States	~5.0	~14%	▼ Declining
India	~3.0	~8%	▲ Rising
EU-27	~2.7	~7%	▼ Declining
Germany	~0.6	~1.7%	▼ Declining
Brazil	~0.5	~1.4%	● Stable
United Kingdom	~0.3	~0.9%	▼ Declining
France	~0.3	~0.8%	▼ Declining

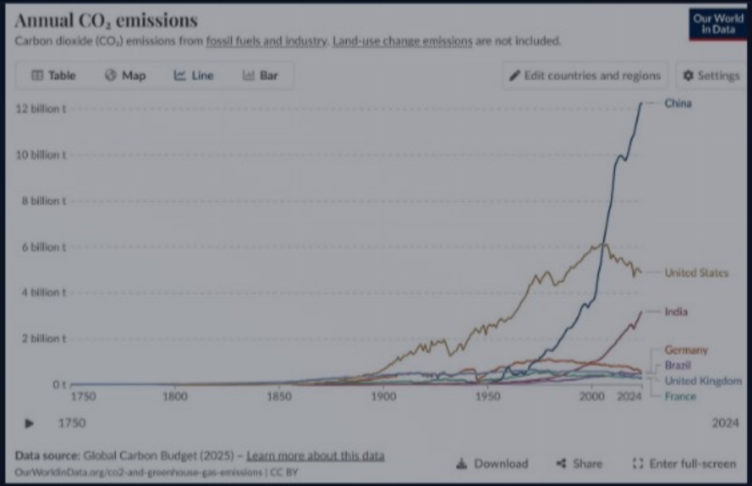


Figure 3: Annual CO_2 emissions by country

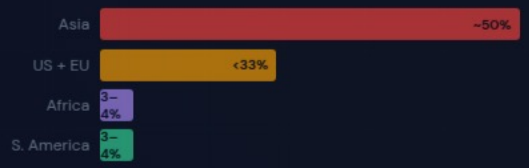
China's steep rise post-2000 contrasts with the US decline from its ~6 Bt peak around 2005.

04 Asia Now ~50%; US + Europe Below One-Third

1900

→

2024



05 Per Capita Gap: 15 t US vs. 0.1 t Chad — a 150× Disparity

Tonnes CO_2 per person per year

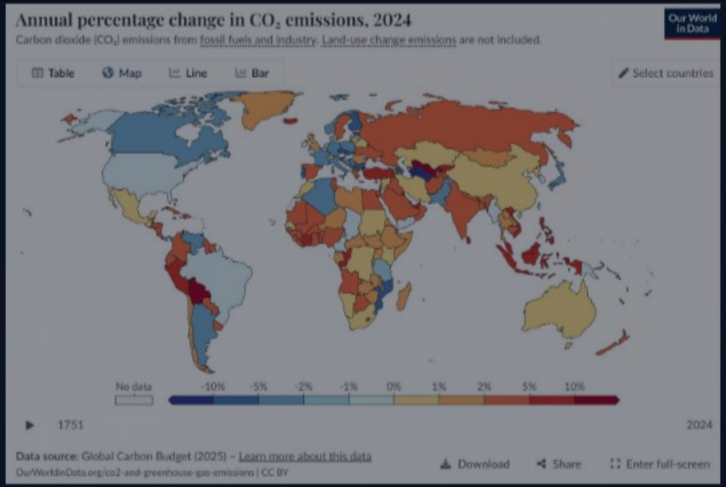


Figure 1: Per capita CO_2 emissions over time

06 2024: US & Europe Cut Emissions; Russia & SE Asia Increase

Source: Global Carbon Budget 2025

07 Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have **very different** per capita emissions depending on energy sources.

COUNTRY	NUCLEAR	RENEWABLES	FOSSIL FUELS	PER CAPITA CO_2
LOWER EMISSIONS — Higher Nuclear & Renewables				
🇫🇷 France	~70%	~12%	~18%	~4 t
🇬🇧 United Kingdom	~15%	~43%	~42%	~4 t
HIGHER EMISSIONS — Higher Fossil Fuel Share				
🇩🇪 Germany	~0%*	~50%	~50%	~7 t
🇳🇱 Netherlands	~3%	~25%	~72%	~8 t

08 Key Takeaways: Emissions Still Rising, But the Map of Responsibility Is Shifting

1. **Global CO_2 emissions exceed 35 billion tonnes per year** and have not yet peaked, despite slowing growth rates.
2. **China alone accounts for over 25%** of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is third-largest and rising fast.
3. The geographic center of emissions has **shifted dramatically**: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.
4. **Per capita inequalities are extreme** — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
5. **Energy mix and policy choices matter**: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on nuclear, renewables, or fossil fuel reliance.

Source: Global Carbon Budget 2025 / NASA / Our World in Data